Chapter 12

Nonlinear Wave Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 12; printed p. 617; PDF page 634.

Chinese reference: 中译本对应章：非线性波动方程，书内第 437 页起。

Editorial reconstruction. We develop energy, causality, local solvability, defocusing global existence, and focusing blow-up. The exposition, expanded arguments, Chinese mathematical notes, problem solutions, and diagrams are reconstructed study material. Source identifiers are recorded separately from our numbering. In particular, a weak energy inequality is not smooth energy conservation, and a noninteger power does not automatically give a smooth nonlinearity. The critical proof states its external linear estimate explicitly.

12.1 Introduction

12.1.1 The equation and its conserved energy

Write $\Box = \partial_t^2 - \Delta_x$. The main equation is

$$\Box u + f(u) = 0, \qquad u(0) = g, \quad u_t(0) = h \quad \text{on } \mathbb{R}^n. \tag{12.1}$$

We also study $\Box u + N(Du, u_t, u) = 0$, with $N(0,0,0) = 0$. Its highest-order part is still the constant wave operator. Although Evans calls it mildly quasilinear, its nonlinearity involves only lower derivatives; equations with solution-dependent principal coefficients require additional analysis. Throughout $D = D_x$. For a scalar f set $F(z) = \int_0^z f(a)\,da$, so $F(0) = 0$.

Theorem 12.1 (Energy; source §12.1.1, Theorem 1)

For a smooth solution with support in one common compact spatial set on each finite time strip, or sufficient decay to justify the same integrations,

$$E(t) = \int_{\mathbb{R}^n} \left\{ \frac{1}{2}(u_t^2 + |Du|^2) + F(u) \right\} dx = E(0). \tag{12.2}$$

非线性项的符号

本章把 f 放在左侧。$f(u) = |u|^{p-1}u$ 给非负势能，$f(u) = -|u|^{p-1}u$ 给负势能。后者才可能满足负能量爆破条件；不能只比较幂次而忽略符号。

Expanded proof (local identity before integration).

Put $e = \frac{1}{2}(u_t^2 + |Du|^2) + F(u)$. The product rule gives

$$e_t - \operatorname{div}(u_t Du) = u_t(\Box u + f(u)) = 0.$$

Integrate on a ball containing the support over a fixed time strip; its boundary term is zero. Under decay hypotheses use cutoffs with radius tending to infinity and check that $u_t Du$ is integrable. This also explains why control of both kinetic and gradient energy is sufficient to eliminate the flux at infinity. For weak solutions the product $u_t \Box u$ may be undefined; conservation must then be established by approximation, not by this formal multiplication. □

12.1.2 Backward cones and the exact flux

For apex (x_0, T) put

$$\begin{aligned} B_t &= B(x_0, T-t), \\ K_{s,\tau} &= \{(x,t) : s < t < \tau,\ x \in B_t\}, \\ \Gamma_{s,\tau} &= \{(x,t) : s < t < \tau,\ |x - x_0| = T - t\}. \end{aligned}$$

On the mantle let $\nu = (x - x_0)/|x - x_0|$. Write $E_K(t) = \int_{B_t} e(x,t)\,dx$.

Theorem 12.2 (Cone flux; source §12.1.2, Theorem 2)
For a solution smooth up to every truncated cone,

$$E_K(s) - E_K(\tau) = \frac{1}{\sqrt{2}} \int_{\Gamma_{s,\tau}} \left\{ \tfrac{1}{2} |u_t \nu - Du|^2 + F(u) \right\} dS. \tag{12.3}$$

If $F \geq 0$, the local cone energy is nonincreasing. If its initial section has $g = h = 0$, then $u = 0$ in that cone.

Expanded proof (moving boundary and the surface factor).

Reynolds' formula for the ball shrinking at unit speed gives

$$E_K'(t) = \int_{\partial B_t} (u_t \partial_\nu u - e)\, dS_x = -\int_{\partial B_t} \{ \tfrac{1}{2} |u_t \nu - Du|^2 + F(u) \}\, dS_x.$$

The square is $u_t^2 + |Du|^2 - 2u_t \partial_\nu u$; its tangential part must not be dropped. Integrating in time proves the result, because the space-time mantle measure satisfies $dS = \sqrt{2}\, dS_x dt$. If the initial local energy is zero, positivity forces $u_t = Du = 0$ in every smaller section. Integrating along vertical segments from the zero initial data gives $u = 0$ there, even if F has other zeroes. When u is smooth at the apex, $E_K(\tau) \to 0$ as $\tau \uparrow T$. We retain this term whenever the apex might be singular: smoothness only for $t < T$ does not justify dropping it. □

锥面上的两种测度

锥面 $t = T - |x - x_0|$ 的诱导测度为 $\sqrt{2}\,dx$，分时切片则为 $\sqrt{2}\,dS_x dt$。因此式中的 $1/\sqrt{2}$ 是几何因子。临界点可能奇异时，顶端剩余能量不能预先设为零。

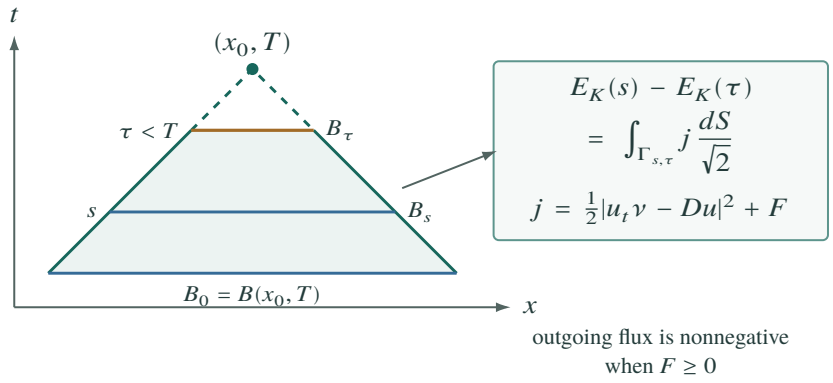


Diagram T12.1. The shrinking sections and the null flux A spatial section of the backward unit-speed cone has radius $T-t$. Energy leaves through the mantle with density $\frac{1}{2}|u_t \nu - Du|^2 + F(u)$, measured using $dS/\sqrt{2}$. The top section at $\tau < T$ is retained. The drawing is a one-space-dimensional section of the n-dimensional cone, not a reduction of the PDE.

12.1.3 Finite propagation without a positive potential

Theorem 12.3 (Domain of dependence; source §12.1.2, Theorem 3)
Let N be locally Lipschitz, $N(0) = 0$, and let two classical solutions of $\Box u + N(Du, u_t, u) = 0$ have bounded first-order state on a compact backward cone. If their two initial data agree on its base, they agree inside the cone. In particular compactly supported data yield support in $B(0, R + t)$ while a classical solution exists.

Expanded proof (difference energy and local Lipschitz constants).

For $w = u - v$, write $Z_u = (Du, u_t, u)$ and define Z_v similarly. The local Lipschitz bound is

$$|N(Z_u) - N(Z_v)| \leq L(|Dw| + |w_t| + |w|),$$

where L depends only on a bound for the two states in this cone. Add $w^2/2$ to the usual wave energy and integrate on its shrinking sections. The lateral boundary contribution is $-\frac{1}{2}|w_t \nu - Dw|^2 - w^2/2 \leq 0$. The interior contribution is $w_t\{w - (N(u) - N(v))\}$ and is at most $C_L(w_t^2 + |Dw|^2 + w^2)$. Thus $Y'(t) \leq C_L Y(t)$ and $Y(0) = 0$; Gronwall gives $Y = 0$. Comparison with zero uses $N(0) = 0$. For energy solutions with globally Lipschitz N, the same inequality is obtained by time/spatial regularization and smooth cone cutoffs, then passage to the limit in L^2. Thus it is also available for the local-data gluing in the next section. A cone is characteristic for the fixed principal wave operator; adding lower-order derivative terms does not change its speed. □

12.2 Existence of Solutions

12.2.1 The linear estimate and the energy solution class

For an integer $k \geq 1$, write

$$\|(u, u_t)\|_{\mathscr{E}_k(I)} = \sup_{t \in I} \{\|u(t)\|_{H^k} + \|u_t(t)\|_{H^{k-1}}\}.$$

The inhomogeneous H^k norm includes the low-frequency L^2 term. We use the whole-space linear wave construction from Chapters 2 and 7: Fourier multipliers $\cos(t|D|)$ and $\sin(t|D|)/|D|$ give, for $u_{tt} - \Delta u = G$,

$$\|(u, u_t)\|_{\mathscr{E}_k([0,T])} \leq C_T \left(\|g\|_{H^k} + \|h\|_{H^{k-1}} + \|G\|_{L^1(0,T;H^{k-1})} \right). \tag{12.4}$$

Here C_T is bounded for $T \leq 1$.

Expanded proof (the energy estimate with its missing low mode).

For smooth data apply D^α, $|\alpha| \leq k-1$, and multiply by $D^\alpha u_t$. The square root of $\|D^\alpha u_t\|_2^2 + \|DD^\alpha u\|_2^2$ grows at most by $\|D^\alpha G\|_2$. Replace this square root by $\sqrt{\varepsilon + \cdots}$ if it vanishes, then let $\varepsilon \downarrow 0$. The remaining $\|u(t)\|_2$ is bounded by $\|g\|_2 + \int_0^t \|u_t(s)\|_2 ds$. Summing gives (12.4); density gives a solution in $C_t H^k \cap C_t^1 H^{k-1}$. Duhamel's formula gives continuity also for $G \in L_t^1 H^{k-1}$. Thus no Poincare constant depending on an expanding spatial ball is needed. □

An energy solution below has $u \in C([0,T]; H^1)$, $u_t \in C([0,T]; L^2)$ and satisfies the PDE in distributions. If $N(Du, u_t, u) \in L^2_{\mathrm{loc}}$, then $u_{tt} \in L^2_{\mathrm{loc}}(0,T; H^{-1}_{\mathrm{loc}})$ and the usual weak formulation against compactly supported H^1 tests follows. The two time traces specify g, h separately.

12.2.2 Globally Lipschitz nonlinearities

Theorem 12.4 (Source §12.2.1, Theorem 1)
Assume $N : \mathbb{R}^{n+2} \to \mathbb{R}$ is globally Lipschitz, $N(0) = 0$. For $g \in H^1$, $h \in L^2$ there is a unique global energy solution of $\Box u + N(Du, u_t, u) = 0$. For $g \in H^2$, $h \in H^1$ it also has $u \in L^\infty(0,T;H^2)$, $u_t \in L^\infty(0,T;H^1)$, $u_{tt} \in L^\infty(0,T;L^2)$ on every finite strip. With $H^1_{\mathrm{loc}}, L^2_{\mathrm{loc}}$ data the corresponding statements hold locally in space, with uniqueness in the local energy class.

Expanded proof (contraction, restart, and local gluing).

On the complete space $C_t H^1 \cap C_t^1 L^2$ with the prescribed initial traces, define $A[v]$ by the linear equation with forcing $-N(Dv, v_t, v)$.

Since $N(0) = 0$, this forcing belongs to $C_t L^2$. For two inputs, (12.4) gives

$$\|A[v] - A[w]\|_{\mathscr{E}_1} \le C_1 LT \|v - w\|_{\mathscr{E}_1}.$$

The homogeneous solution makes the affine space nonempty. Choose $T \le 1$ so $C_1 LT < 1/2$ and apply Banach's theorem. The same step size works for every new initial state, so repeated restart constructs a solution on every finite interval. Its norm is finite there, for example by the integral estimate and Gronwall; no finite accumulation of restart times occurs.

For the additional derivative, apply the spatial difference quotient δ_ε to the PDE. Pointwise Lipschitz continuity gives

$$|\delta_\varepsilon N(Du, u_t, u)| \le L(|D\delta_\varepsilon u| + |\delta_\varepsilon u_t| + |\delta_\varepsilon u|).$$

The difference energy and Gronwall bound the quotient in $L^\infty_t H^1$ and its time derivative in $L^\infty_t L^2$, uniformly in ε, by the $H^2 \times H^1$ initial norm. Weak compactness and the difference-quotient characterization give $D^2 u, Du_t \in L^\infty_t L^2$; the equation then gives the asserted bound for u_{tt}. Differentiability of a merely Lipschitz N is not needed for this argument.

For local data, cut them off outside $B(0, j)$ and solve the global problem. Two cutoff solutions coincide in the backward domains where their data agree, by the regularized difference-cone estimate from §12.1. For every fixed compact space-time cylinder they therefore agree for all sufficiently large j. Their common values define a solution independent of the cutoff. The same cone argument proves uniqueness and localizes the higher-regularity estimates. □

12.2.3 Sobolev products and composition

Theorem 12.5 (Source §12.2.2, Theorem 2; Problem 12)
Let $r > n/2$ be an integer, $u_1, \dots, u_m \in H^r(\mathbb{R}^n)$. For multiindices with $\sum_j |\beta_j| \le r$,

$$\left\| \prod_j D^{\beta_j} u_j \right\|_2 \le C_{n,r,m} \prod_j \|u_j\|_{H^r}. \tag{12.5}$$

If $P : \mathbb{R}^m \to \mathbb{R}$ is smooth and $P(0) = 0$, then $P(u) \in H^r$, with a bound depending on the H^r norms and derivatives of P on the resulting bounded range.

Expanded proof (including the exponent allocation).

Put $b_j = |\beta_j|$. Sobolev embedding gives $D^{\beta_j} u_j \in L^{2r/b_j}$ for $0 < b_j \le r$ and $u_j \in L^\infty$ for $b_j = 0$. Indeed for $b_j < r$ the choice $1/p = b_j/(2r)$ lies

原点条件与全空间

若 $N(0) \neq 0$，零状态也会受到常量强迫，$N(Dv, v_t, v)$ 不一定在 $L^2(\mathbb{R}^n)$。$N(0) = 0$ 同时保证零解、有限传播中的比较基准与全空间能量构造。

strictly above $1/2 - (r - b_j)/n$, since $2r > n$; $b_j = r$ gives $p = 2$. Each derivative also lies in L^2. By interpolation choose $1/p_j \in [b_j/(2r), 1/2]$ so that $\sum_j 1/p_j = 1/2$. This is possible because the sum of the lower endpoints is at most $1/2$ and that of the upper endpoints is at least $1/2$. Holder's inequality proves (12.5). On the infinite measure space one must make this sum exact, rather than infer an L^2 estimate from a product lying only in a larger L^p space.

For $0 < |\alpha| \le r$, the chain rule expresses $D^\alpha P(u)$ as a finite sum of bounded coefficients $D^j P(u)$ times products of derivatives of the u_i with total order $|\alpha|$. Apply (12.5). For the undifferentiated term use $P(0) = 0$ and $|P(u)| \le C|u|$ on the bounded range. Approximation by smooth functions proves the Sobolev chain rule and the assertion for all H^r inputs. The same estimates applied to $P(u) - P(v) = \int_0^1 DP(v + \theta(u - v))(u - v)\,d\theta$ show local Lipschitz dependence in H^r; subtract the constant $DP(0)$ before using the algebra estimate on its coefficient. □

12.2.4 Local existence and the role of the two norms

Theorem 12.6 (Source §12.2.2, Theorem 3)
Assume N is smooth with $N(0) = 0$, and $k > n/2 + 1$ is an integer. For $g \in H^k$, $h \in H^{k-1}$ there are $T > 0$ and a unique solution

$$u \in C([0,T]; H^k), \qquad u_t \in C([0,T]; H^{k-1}).$$

The lifetime depends on these norms and the nonlinearity on a bounded state range. Classical C^2 regularity follows if $k > n/2 + 2$.

Expanded proof (closed strong ball with the weaker metric).

Fix $T \le 1$ and let Y_R consist of pairs (v, v_t) continuous in $H^1 \times L^2$, with the prescribed traces, and with $\|(v, v_t)\|_{\mathscr{E}_k} \le R$ in the essential-supremum sense. Use the $\mathscr{E}_1$ metric. This space is complete: a Cauchy sequence has a strong low-norm limit; the high-norm bounded sequence has a weak-star subsequence in $L^\infty(H^k \times H^{k-1})$. Both limits agree as distributions. Weak lower semicontinuity gives the bound R, and low-norm convergence preserves the derivative relation and both initial traces. The uniform high bound can also be assigned at each time by weak limits of nearby good time slices.

For $A[v]$ as above, composition at order $k - 1 > n/2$ gives

$$\|N(Dv, v_t, v)\|_{H^{k-1}} \le \Phi(R).$$

Take $R > 2C_1(\|g\|_{H^k} + \|h\|_{H^{k-1}}) + 1$. The free wave belongs to Y_R, and

$$\|A[v]\|_{\mathscr{E}_k} \le C_1 D_0 + C_1 T \Phi(R) \le R$$

两个范数各做什么

高阶界使 (Du, u_t, u) 的值域有界，从而给出 N 的局部 Lipschitz 常数。低阶能量范数用于收缩。只说迭代在弱范数收敛，还需证明强范数球在该度量下闭合。

for small T. Sobolev embedding bounds all first-order states by CR, so the local Lipschitz constant L_R gives $\|A[v]-A[w]\|_{\mathscr{E}_1} \le C_1 T L_R \|v-w\|_{\mathscr{E}_1}$. Shorten T to make this less than one half of the input distance. Banach's theorem gives a fixed point in the closed space Y_R. The resulting forcing is in $L^1_t H^{k-1}$; linear uniqueness and (12.4) then upgrade the solution to the asserted strong time continuity. Differences of two bounded high-regularity solutions obey the same low-energy Gronwall estimate, proving uniqueness without restricting them to one preselected ball. If $k > n/2+2$, D^2u, Du_t, u_{tt} are continuous by embedding and the PDE. □

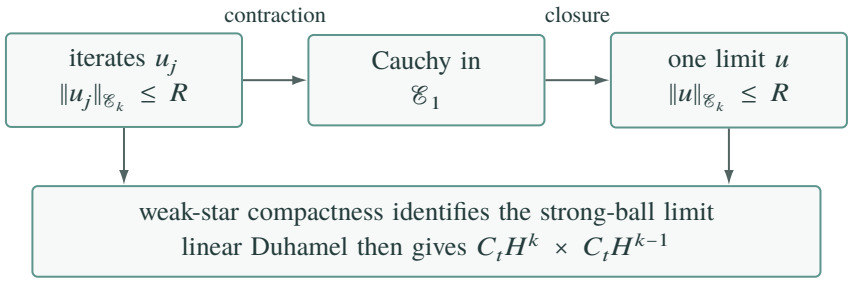


Diagram T12.2. A complete ball for the weak-metric iteration Every iterate stays in a bounded high Sobolev ball. Contraction is measured in the lower energy norm; weak-star compactness identifies its limit with the low-norm limit, preserving the high bound. The final linear equation restores strong time continuity. The arrows describe a proof, not a numerical convergence rate in the high norm.

12.3 Semilinear Wave Equations

12.3.1 A sign condition gives weak existence

Assume f is smooth and $zf(z) \ge 0$. Then $F \ge 0$. The resulting energy controls Du, u_t, but it does not directly control $\|u\|_2$ uniformly for all positive time.

Theorem 12.7 (Sign condition; source §12.3.1, Theorem 1)
If $g \in H^1$, $h \in L^2$ and $F(g) \in L^1$, there is a global distributional solution with $f(u) \in L^1_{\mathrm{loc}}$ and

$$\|Du(t)\|_2^2 + \|u_t(t)\|_2^2 + 2\int F(u(t)) \le 2E(0).$$

One can choose $u \in C([0,\infty); L^2)$ with $\|u(t)\|_2 \le \|g\|_2 + t\sqrt{2E(0)}$, and Du, u_t weakly continuous into L^2. The energy inequality holds at every time for these representatives. The initial energy traces are strong. No uniqueness or energy equality is asserted in this general class.

能量不控制全时间质量

这里 F 可为零，能量只控制 Du, u_t。由 $u(t) = g + \int_0^t u_t$ 得到的是 $\|u(t)\|_2 \le \|g\|_2 + t\sqrt{2E(0)}$。因此一般只能写每个有限时间条带上的 $L^\infty_t H^1_x$ 界。

Expanded proof (truncation, uniform integrability, and time slices).

Choose even smooth cutoffs $0 \le \chi_j \le 1$, equal to one on $[-j,j]$ and zero outside $[-2j,2j]$. Put $f_j = \chi_j f$ and $F_j(z) = \int_0^z f_j(a)da$. Then f_j is globally Lipschitz, $zf_j(z) \ge 0$, $0 \le F_j \le F$, and $F_j = F$ on $[-j,j]$. The previous section gives an energy solution u_j. Conservation for this Lipschitz problem follows by approximating the data with smooth data in $H^1 \times L^2$ and passing through the linear energy estimate; $F_j(z) \le C_j z^2$ makes the potential continuous in L^1 under L^2 convergence on bounded energy sets. Consequently $E_j(t) = E_j(0) \le E(0)$ and

$$\|Du_j(t)\|_2 + \|(u_j)_t(t)\|_2 \le C_E, \qquad \|u_j(t)\|_2 \le \|g\|_2 + tC_E.$$

Dominated convergence also gives $E_j(0) \to E(0)$. Rellich compactness on bounded balls and the equicontinuity $\|u_j(t) - u_j(s)\|_2 \le C_E|t-s|$ give, by a diagonal Arzela–Ascoli argument, $u_j \to u$ in $C([0,T]; L^2_{\mathrm{loc}})$ for every finite T, and a.e. in space-time along a subsequence. The first derivatives converge weak-star in $L^\infty_t L^2_x$.

Multiplying the approximate PDE by u_j and integrating yields

$$\int_0^T \int u_j f_j(u_j) = \int_0^T \int ((u_j)_t^2 - |Du_j|^2) - \Big[\int u_j (u_j)_t\Big]_0^T \le C_{E,g,T}.$$

All terms are justified first for smooth data and then by the same energy approximation. The integrand on the left is nonnegative. For a measurable subset A of a compact space-time cylinder and $M > 0$,

$$\int_A |f_j(u_j)| \le C_{E,g,T}/M + |A| \max_{|z|\le M} |f(z)|.$$

The first term comes from $|u_j| > M$; the second uses $|f_j| \le |f|$. This is uniform integrability. Also $f_j(u_j) \to f(u)$ a.e., because at any point with finite limit u, eventually $|u_j| < j$. Egoroff on the compact cylinder, followed by the preceding small-set bound, proves $f_j(u_j) \to f(u)$ in L^1 there. This is a space-time L^1_{loc} statement, not a claim of convergence at every fixed time. The weak equation now passes to the limit. For a smooth test reaching $t = 0$ its first-order form is

$$\iint \{-u_t\varphi_t + Du \cdot D\varphi + f(u)\varphi\} = \int h\varphi(\cdot,0), \qquad u(0) = g.$$

To obtain time representatives, for each compactly supported smooth ψ use $\frac{d}{dt}\int (u_j)_t\psi = \int u_j\Delta\psi - f_j(u_j)\psi$. Uniform integrability makes these scalar functions equicontinuous, including at zero. Diagonal extraction for a dense collection of ψ, and the uniform L^2 bounds, give weak L^2 convergence of the velocities at every time and weak time continuity of u_t. The gradients converge weakly at every time by the strong local convergence of u_j. At a fixed time, extract a further subsequence with a.e. spatial

convergence on expanding balls. Fatou for $F_j(u_j)$ and weak lower semicontinuity of the two quadratic terms give $E(t) \le E(0)$, independently of the chosen subsequence. Thus this inequality holds for the chosen representatives at all times. The identity $u(t) = g + \int_0^t u_t(s)ds$ gives the asserted global L^2 continuity and growth bound. As $t \downarrow 0$, lower semicontinuity of the potential and weak convergence of (Du, u_t), together with $E(t) \le E(0)$, force convergence of their squared norms; weak convergence plus convergence of Hilbert norms is strong. This proves both strong initial energy traces. A diagonal extraction in $T = 1, 2, \dots$ completes global existence. □

12.3.2 Three-dimensional retarded potentials and continuation

For smooth compactly supported g, h let v be their free wave. In three dimensions Duhamel's formula is

$$u(x,t) = v(x,t) - \frac{1}{4\pi} \int_{B(x,t)} \frac{f(u(y, t-|y-x|))}{|y-x|}\, dy. \tag{12.6}$$

Theorem 12.8 (Source §12.3.2, Theorem 2)
If f is smooth, $f(0) = 0$, this problem has a unique local smooth solution. If its maximal smooth lifetime $T_* < \infty$, then

$$\sup_{0 \le t < T_*} \|u(t)\|_\infty = \infty.$$

Equivalently, a bound for u on a finite strip permits continuation. We use this strip-bound criterion rather than infer a pointwise limit from the mere failure of a uniform bound.

Expanded proof (the kernel norm and derivative persistence).

On $0 \le t \le T \le 1$, take the closed sup-norm ball $\|u - v\|_\infty \le 1$ of bounded continuous functions with initial trace g. Its range is in $[-M, M]$ for a fixed M. The right side of (12.6) defines a map with

$$\|A[u] - A[w]\|_\infty \le \tfrac{1}{2} L_M T^2 \|u - w\|_\infty, \qquad \|A[u] - v\|_\infty \le \tfrac{1}{2} T^2 \max_{[-M,M]} |f|,$$

because $(4\pi)^{-1} \int_{B(0,T)} |y|^{-1} dy = T^2/2$. It is a contraction of the ball for small T.

The fixed point is smooth. For instance spatial difference quotients satisfy a linear retarded equation with coefficient $\int_0^1 f'(\theta u(x + \varepsilon e_j, t) + (1-\theta)u(x,t))d\theta$. It is uniformly bounded on a bounded range. The retarded operator has the estimate $\|R[G](t)\|_\infty \le \int_0^t (t-s)\|G(s)\|_\infty ds$. Gronwall therefore bounds the first spatial derivatives. The time derivative satisfies the differentiated wave equation with initial data $(h, \Delta g - f(g))$

and is bounded in the same way. Successive differentiation gives an equation with leading term $f'(u)D^\alpha u$ and forcing made from lower derivatives, already bounded by induction. Difference quotients justify each differentiation and convergence; the corresponding linear retarded equation has a unique continuous solution. This bounds every derivative on a finite strip if u is bounded there. Finite propagation provides a common support ball. Uniform bounds on one further derivative give limiting smooth data at T_*; applying local existence to them extends the solution, a contradiction. Alternatively higher-order energy limits provide the same restart. Uniqueness follows either from contraction on successive short strips or the difference-cone energy estimate. □

12.3.3 Subcritical defocusing powers

Consider $f(u) = |u|^{p-1}u$, $1 < p < 5$, in $n = 3$.

Lemma 12.9 (Cone-trace bounds; expanded source §12.3.3)
For a smooth defocusing solution and $t \le T_0 < \infty$, put $w(y) = u(y, t-|y-x|)$ on $B(x,t)$. Then

$$\|Dw\|_2 + t^{-1}\|w\|_2 + \|w/|y-x|\|_2 + \|w\|_6 \le C(E(0)^{1/2} + T_0^{1/2}\|g\|_\infty). \tag{12.7}$$

Constants are independent of the apex and the cone radius.

Expanded proof (the nonzero boundary value in Hardy's inequality).

The chain rule gives $Dw = Du - \nu u_t$ on the mantle. Thus (12.3) gives $\int |Dw|^2 \le 2E(0)$. Let $\overline{w}$ be the ball average. Integrating $|u_t|$ along vertical lines in the cone gives

$$\int_{B(x,t)} |w| \le \int_{B(x,t)} |g| + \iint_{K_{0,t}} |u_t|.$$

Indeed each vertical segment from time zero to $t - |y-x|$ lies in the cone. Since its space-time volume is Ct^4 and $\iint_K u_t^2 \le 2tE(0)$, $|\overline{w}| \le Ct^{-1/2}E(0)^{1/2} + \|g\|_\infty$. Poincare's inequality yields $t^{-2}\|w\|_2^2 \le C\|Dw\|_2^2 + Ct|\overline{w}|^2$. The scaled ball inequalities are

$$\|w\|_6 + \|w/|y-x|\|_2 \le C(\|Dw\|_2 + t^{-1}\|w\|_2).$$

They follow by scaling the unit-ball Sobolev and Hardy inequalities. For the latter, extend the unit-ball function by a bounded H^1 extension and cutoff to $\mathbb{R}^3$, then apply whole-space Hardy. One cannot apply the zero-trace version directly to this cone trace. These inequalities and the average bound prove the lemma. □

实数幂与光滑性

当 p 不是正奇整数，$|z|^{p-1}z$ 在零点通常不是 C^∞。下面对所有 $1 < p < 5$ 声称有限能量有界 mild 解；只有非线性本身光滑时，才由导数持续性得到光滑解。

Theorem 12.10 (Subcritical existence; source §12.3.3, Theorem 3)
For compactly supported smooth data and $1 < p < 5$, there is a unique global finite-energy mild solution, bounded on each finite time strip, with $u \in C_t H^1$, $u_t \in C_t L^2$ and conserved energy. When the nonlinearity is smooth, the solution is smooth.

Expanded proof (the sublinear bootstrap and the power qualification).

First work in a smooth regime. Formula (12.6) and Cauchy–Schwarz give

$$|u(x,t)| \le \|v\|_\infty + C\|w/|y-x|\|_2 \left(\int_{B(x,t)} |w|^{2(p-1)} dy \right)^{1/2}.$$

For $1 < p \le 4$, the last exponent is at most six. Holder on the ball and (12.7) give a uniform bound for $t \le T_0$; this uses integrals, not an invalid norm triangle inequality when $2(p-1) < 1$. For $4 < p < 5$, put $M(\tau) = \sup_{x, 0 \le t \le \tau} |u(x,t)|$. Then

$$\left(\int |w|^{2(p-1)} \right)^{1/2} \le M(\tau)^{p-4} \|w\|_6^3, \quad M(\tau) \le C_0 + C_1 M(\tau)^{p-4}.$$

Since $0 < p - 4 < 1$, Young's inequality absorbs the last term into $M/2$ plus a constant. Thus M is bounded independently of the current smooth lifespan below T_0. The continuation theorem proves global smooth existence whenever f is smooth.

For an arbitrary real p, approximate f by smooth sign-preserving nonlinearities, cut off at large amplitudes, converging in C^1 on every bounded interval and with $|f_j(z)| \le C(|z| + |z|^p)$. For example smooth $z(z^2 + \varepsilon_j^2)^{(p-1)/2}$ first and then truncate far out. Their nonnegative potentials and compact initial data have uniformly bounded energy. The preceding argument gives a uniform strip bound; the extra linear term is bounded by the same Hardy estimate times the square root of the ball volume. On this bounded range the nonlinearities have one uniform Lipschitz constant. Difference energy estimates, using uniform convergence of f_j on that range and the common finite support ball, show Cauchy convergence in $C_t H^1 \times C_t L^2$. The retarded Volterra difference estimate also gives Cauchy convergence in the supremum norm. The retarded formula passes to the limit, and its sup-norm contraction gives the same mild solution. The potentials converge uniformly on the bounded range, so their integrals converge and energy equality passes to the limit. The difference estimate also proves uniqueness among bounded energy solutions on each finite strip. Smoothness across zeros is not claimed for the noninteger powers. This supplies the regularity qualification absent from a literal all-power smoothness statement. □

12.4 Critical Power Nonlinearity

12.4.1 Scaling and the external linear estimate

In $\mathbb{R}^3$ the defocusing equation

$$\Box u + u^5 = 0, \qquad E = \int \{(u_t^2 + |Du|^2)/2 + u^6/6\} \tag{12.8}$$

is energy critical. For the general power, $u_\lambda(x,t) = \lambda^{2/(p-1)} u(\lambda x, \lambda t)$ solves the same equation, while its energy scales by $\lambda^{4/(p-1)-1}$. At $p = 5$ this factor is one: a small spatial scale cannot be excluded by total energy alone.

Theorem 12.11 (Linear estimate; source §12.4, Theorem 1)
For $\Box v = G$ on a time interval $I = [s,\tau]$ in $\mathbb{R}^3$, assume $v(s) \in \dot H^1$, $v_t(s) \in L^2$, and $G \in L^1(I;L^2)$. Then

$$\begin{aligned} &\|Dv, v_t\|_{L^\infty(I;L^2)} + \|v\|_{L^\infty(I;L^6)} + \|v\|_{L^4(I;L^{12})} \\ &\quad \le C\{\|Dv(s)\|_2 + \|v_t(s)\|_2 + \|G\|_{L^1(I;L^2)}\}. \end{aligned} \tag{12.9}$$

We explicitly use the homogeneous three-dimensional Strichartz estimate for the $(4,12)$ pair as an external theorem*, with constant independent of the interval. It is not proved here.*

Expanded proof (the proved energy part and the external input).

The energy argument of §12.2, without the L^2 position term, gives the first norm on the left. The homogeneous Sobolev inequality $\|v\|_6 \le C\|Dv\|_2$ gives the second. The external homogeneous estimate bounds the third norm for free waves by their $\dot H^1 \times L^2$ data. Apply this to the free evolution starting at each forcing time and use Minkowski in Duhamel's integral. Its $\dot H^1 \times L^2$ size is $\|G(t)\|_2$, giving the forcing term in (12.9). Restriction of the homogeneous estimate on the whole time line explains uniformity under interval shortening. The dispersive proof of that homogeneous estimate remains external; energy conservation alone does not imply an $L_t^4 L_x^{12}$ bound. □

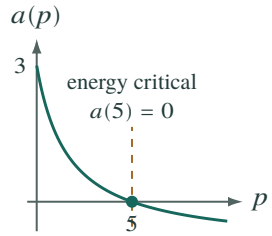
12.4.2 Morawetz identity and potential nonconcentration

Fix an apex (x_0, T) before which u is smooth and has finite conserved energy. Set $x_0 = 0$ by translation and $r = |x|$. The local energy $E_K(t)$ decreases, so it has a limit $E_* \ge 0$ as $t \uparrow T$. Define the tail flux

$$\phi(s) = E_K(s) - E_* = \frac{1}{\sqrt{2}} \int_{\Gamma_{s,T}} \{\tfrac{1}{2}|u_t \nu - Du|^2 + u^6/6\}\, dS. \tag{12.10}$$

Monotone convergence in the truncated flux identity gives this formula and $\phi(s) \to 0$. At this stage $E_* = 0$ is *not* known.

图 T12.M：能量临界的缩放指数



图中 $a(p) = 4/(p-1) - 1$，且 $E[u_\lambda] = \lambda^{a(p)} E[u]$。$p = 5$ 是三维退聚焦波方程的能量临界指数；它与后面聚焦小数据爆破的 $1 + \sqrt{2}$ 阈值属于不同问题。

Lemma 12.12 (Multiplier identity; source equations (6)–(8))
Put $\theta = t - T$, $m = \theta u_t + x \cdot Du + u$ *and*

$$P = \theta\{(u_t^2 + |Du|^2)/2 + u^6/6\} + (x \cdot Du + u)u_t,$$

$$Q = mDu + x\{(u_t^2 - |Du|^2)/2 - u^6/6\}.$$

Then $P_t - \operatorname{div} Q = m(\Box u + u^5) - u^6/3$.

Expanded proof (an identity also valid off solutions).

Use $\theta_t = 1$, $\operatorname{div} x = 3$ and $D(x \cdot Du) = Du + D^2u\,x$. The mixed terms $\theta Du \cdot Du_t$ cancel, as do $x \cdot D(u_t)u_t$ and $x \cdot D^2u\,Du$ after differentiation. The remaining second derivatives form $m(u_{tt} - \Delta u)$. The potential terms are $mu^5 - u^6/3$: their residual coefficient is $1/6 + 3/6 - 1 = -1/3$. The coefficients of u_t^2 and $|Du|^2$ are both zero. This proves the identity without presupposing the PDE. □

先证明通量尾部趋零

正确的关系是 $\phi(s) = E_K(s) - \lim_{t\uparrow T} E_K(t)$。单调性保证尾部趋零，不要求顶端剩余能量预先为零。后面通过线性 Duhamel 的强能量极限再证明全部能量不集中。

Theorem 12.13 (Potential nonconcentration; source §12.4, Theorem 2)
For $R = T - s$,

$$\int_{B(x_0,R)} |u(x,s)|^6 dx \le C\{\phi(s) + \phi(s)^{1/3}\} \longrightarrow 0. \tag{12.11}$$

Since $\phi \le E$, *the right side can also be written* $C_E\phi(s)^{1/3}$, *making its energy dependence explicit.*

Expanded proof (top term, boundary cancellation, and the endpoint).

Translate s to zero, so R replaces T. Integrate the multiplier identity on $K_{0,\tau}$ and let $\tau \uparrow R$. The top integral tends to zero even if the apex is singular: with $\rho = R - \tau$,

$$\left|\int_{B(0,\rho)} P(x,\tau)dx\right| \le C\rho E + C\|u\|_{L^2(B_\rho)}\|u_t\|_2 \le C\rho(E + E^{2/3}).$$

Here Holder gives $\|u\|_{L^2(B_\rho)} \le C\rho\|u\|_6$. Thus, putting $A = \frac{1}{\sqrt{2}}\int_\Gamma (P - Q \cdot \nu)dS$ and $B = \int_{B_R} P(x,0)dx$, we obtain $A - B = -\iint_K u^6/3 \le 0$.

On the mantle $r = R - t$, $P - Q \cdot \nu = -r(u_t - u_r)^2 + u(u_t - u_r)$. Set $w(y) = u(y, R - |y|)$. Then $w_r = u_r - u_t$ and

$$A = -\int_{B_R} r\left(w_r + \frac{w}{r}\right)^2 dy + \frac{1}{2}\int_{\partial B_R} u(x,0)^2 dS_x. \tag{12.12}$$

Indeed $A = -\int (rw_r^2 + ww_r)$ and $\int (w^2/r + ww_r) = \frac{1}{2}\int_{\partial B_R} w^2$, from $\operatorname{div}(y/r) = 2/r$. This calculation is valid for the possibly singular cone trace: flux gives $Dw \in L^2$ and $w \in L^6$, so $w \in H^1(B_R)$ (the point at

zero is removable for H^1 in dimension three). The weighted term w^2/r is integrable by Holder. Approximate in H^1 with the outer smooth trace fixed, or use radial cutoffs and a vanishing inner-boundary subsequence, to justify the integration.

For the bottom term use Young's inequality and $r \le R$:

$$\begin{aligned} B &\le -\frac{R}{6}\int_{B_R} u^6 + \frac{R}{2}\int_{B_R}(|Du + xu/r^2|^2 - |Du|^2) \\ &= -\frac{R}{6}\int_{B_R} u^6 + \frac{1}{2}\int_{\partial B_R} u^2. \end{aligned}$$

The last equality follows from $\operatorname{div}(x/r^2) = 1/r^2$; the origin contributes no boundary term for the smooth bottom data. The boundary terms cancel between this bound and (12.12). Therefore

$$R\int_{B_R} u(x,0)^6 dx \le C\int_{B_R} r(w_r + w/r)^2 \le CR\phi(0) + C\int_{B_R} w^2/r.$$

Finally

$$\int_{B_R} w^2/r \le \left(\int_{B_R} w^6\right)^{1/3}\left(\int_{B_R} r^{-3/2}\right)^{2/3} \le CR\phi(0)^{1/3}.$$

Divide by R and undo the time translation. Every estimate is on truncated smooth cones before a justified limit, so no regularity at the candidate blow-up point has been assumed. □

12.4.3 Localizing Strichartz and closing the global proof

Lemma 12.14 (Localized estimate and the bootstrap barrier)
Let $C = \{s < t < \tau,\ |x - x_0| < R - (t - s)\}$ be a backward cone. For a smooth quintic solution set $X(\tau) = \|u\|_{L^4(s,\tau;L^{12}(C_t))}$. Then

$$X(\tau) \le C_0 + C\sup_{s<t<\tau}\|u(t)\|_{L^6(C_t)} X(\tau)^4, \qquad C_0 \le C\sqrt{E}. \quad (12.13)$$

If the displayed L^6 coefficient is sufficiently small depending only on E, then $X(\tau)$ stays bounded up to the cone apex.

Expanded proof (no unsupported restriction of a global estimate).

Solve the linear equation with data $(u(s), u_t(s))$ on all space and forcing $-\mathbf{1}_C u^5$, zero outside C. Finite propagation for the linear equation implies its solution equals u inside C: the backward cone of any point of C remains inside C down to s. Apply (12.9) to this linear solution. Holder in space and time gives

$$\|u^5\|_{L^1L^2(C)} \le \sup_t \|u(t)\|_{L^6(C_t)}\int_s^\tau \|u(t)\|^4_{L^{12}(C_t)} dt.$$

This proves the estimate using a globally defined forcing and data; no boundary Strichartz theorem is being assumed. Choose $B = 2\max(C_0, 1)$ and require the nonlinear coefficient to be less than $1/(4B^3)$. The continuous function X starts at zero. At a first crossing of B, the inequality would give $B \le C_0 + B/4 \le 3B/4$, impossible. It therefore remains below B. The small coefficient alone is insufficient unless this continuity argument excludes the large branch of $X \le C_0 + \varepsilon X^4$. □

Theorem 12.15 (Global smooth quintic solution; source §12.4, Theorem 3)
For $g, h \in C_c^\infty(\mathbb{R}^3)$, the defocusing quintic problem has a unique global smooth solution. The proof uses the stated external homogeneous Strichartz estimate.

Expanded proof (from potential flux to an actual continuation bound).

Assume a finite maximal time T. For each x_0, potential nonconcentration makes the L^6 coefficient in (12.13) small on a sufficiently short tail of the cone with apex (x_0, T). Thus $X(T) < \infty$ on that cone, and $G = -\mathbf{1}_K u^5 \in L^1(s, T; L^2)$. The linear solution with this forcing and full data at s agrees with u inside the cone, as in the lemma. Because its forcing is integrable in time, Duhamel gives a strong $\dot{H}^1 \times L^2$ limit at T. Strong L^2 convergence of its gradient and velocity, together with absolute continuity of their squared integrals at the limiting time, now implies

$$\int_{B(x_0, T-t)} (|Du|^2 + u_t^2)\,dx \longrightarrow 0.$$

Together with (12.11) this proves $E_K(t) \to 0$. This argument replaces an unproved differentiation estimate on a cone with a characteristic boundary.

Fix x_0. Choose $s < T$ so its local energy is smaller than a chosen $\varepsilon > 0$. At this fixed smooth time, absolute continuity in the radius allows a $\delta > 0$ such that the energy on $B(x_0, T - s + \delta)$ is less than 2ε. Positive cone flux propagates this bound into the enlarged cone $|x - x_0| < T - t + \delta$. Its L^6 norm is at most $(12\varepsilon)^{1/6}$. The localized lemma, still using the full initial energy E for C_0, gives a finite L^4L^{12} norm on this enlarged cone. Choose ε small enough for the fixed barrier.

Finite propagation bounds all supports for $t < T$ by the fixed closed ball $\overline{B}(0, R_0 + T)$. Cover that ball by finitely many of the balls $B(x_j, \delta_j/2)$ just constructed. These balls lie inside the corresponding enlarged sections for every $t < T$. Start after the largest of their finitely many start times, and use the triangle inequality in L^{12} and then L^4. Adding the earlier smooth strip proves $u \in L^4(0, T; L^{12}(\mathbb{R}^3))$. No object $\operatorname{spt} u(\cdot, T)$ is needed before existence at T is known.

Finally $v = D_j u$ solves $\Box v = -5u^4 v$. The homogeneous energy estimate and Sobolev give, for $A_j(t) = \|Dv(t)\|_2 + \|v_t(t)\|_2$,

$$A_j(t) \le C_j + C\int_0^t \|u(s)\|_{12}^4 A_j(s)\,ds, \qquad \|u^4 v\|_2 \le \|u\|_{12}^4 \|v\|_6.$$

Gronwall bounds A_j up to T. Thus Du is bounded in L^6; energy already bounds u in L^6. The embedding $W^{1,6}(\mathbb{R}^3) \hookrightarrow L^\infty$ supplies a uniform amplitude bound. The continuation theorem of §12.3 contradicts maximality. Local uniqueness extends to all time. □

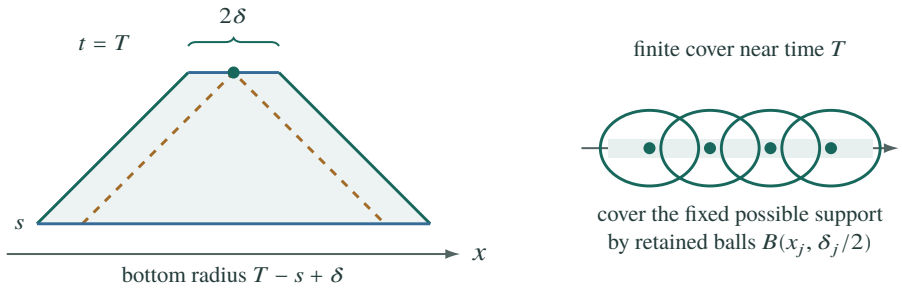


Diagram T12.3. Why the cone must be enlarged The flux tail first makes the potential energy small on shrinking cones. Localized Strichartz and a linear Duhamel limit then exclude concentration of the kinetic and gradient energy. At a fixed time s, the small-energy ball can be enlarged by $\delta > 0$. The enlarged sections retain a ball of radius δ near T, allowing a finite cover of the fixed possible support. A cone with zero terminal radius alone would not provide that cover.

12.5 Nonexistence of Solutions

12.5.1 Negative energy and the concavity method

Consider a classical solution of (12.1) with compactly supported smooth data and sufficient regularity for the following identities. Finite propagation bounds its support on every finite strip. This section allows a negative potential.

Theorem 12.16 (Source §12.5.1, Theorem 1)
Suppose $f(0) = 0$, $F' = f$, $F(0) = 0$, and for some $\lambda > 2$,

$$zf(z) \le \lambda F(z) \quad (z \in \mathbb{R}).$$

If the conserved energy $E(0) < 0$, no such solution can persist for all $t \ge 0$. In particular, the concavity argument forces breakdown in a finite time.

Expanded proof (obtaining a decreasing concave quantity).

Put $I(t) = \frac{1}{2}\int u^2$. Differentiating twice and integrating by parts gives

$$I' = \int uu_t, \qquad I'' = \int (u_t^2 - |Du|^2 - uf(u)).$$

Set $\alpha = (\lambda - 2)/4 > 0$ and $\beta = -\lambda E(0) > 0$. Energy conservation and the structural assumption yield

$$I'' \geq (2 + 2\alpha)\int u_t^2 + 2\alpha \int |Du|^2 + \beta, \qquad (I')^2 \leq 2I\int u_t^2.$$

Therefore $II'' - (1+\alpha)(I')^2 \geq \beta I$. The function I never vanishes: if $u = 0$ on a time slice, its energy there would be $\frac{1}{2}\int u_t^2 \geq 0$. For $J = I^{-\alpha} > 0$,

$$J'' = -\alpha I^{-\alpha-2}\{II'' - (1+\alpha)(I')^2\} \leq -\alpha\beta I^{-\alpha-1} < 0. \tag{12.14}$$

If a global solution existed, $I'' \geq \beta$ would make I' positive at some finite t_0. Then $J'(t_0) < 0$, and concavity implies $J(t) \leq J(t_0) + (t - t_0)J'(t_0)$ for $t \geq t_0$. The right side becomes zero in finite time, contradicting $J > 0$. No positivity assumption on the initial correlation $\int gh$ is needed; $I'' \geq \beta$ creates the required sign later. □

12.5.2 Small-data blow-up below the three-dimensional threshold

Now consider the different focusing equation

$$\Box u = |u|^p \quad \text{in } \mathbb{R}^3, \qquad \int g > 0, \quad \int h > 0, \quad \operatorname{spt}(g,h) \subset B(0,R_0). \tag{12.15}$$

The data themselves need not be nonnegative. For a noninteger p the statement concerns a classical solution with the regularity needed for the representation and integrations, not an automatic C^∞ solution of a nonsmooth nonlinearity.

> **Theorem 12.17** (Source §12.5.2, Theorem 2)
> *If $1 < p < 1 + \sqrt{2}$, the solution in (12.15) cannot exist classically for all positive time. The assumptions allow arbitrarily small positive multiples of fixed admissible data. The endpoint $p = 1 + \sqrt{2}$ is not established by this proof.*

Expanded proof (the shell gain and a divergent time integral).

Assume a global solution and put $I(t) = \int u(x,t)dx$. Finite propagation and integration of the PDE give $I'' = \int |u|^p$, hence $I, I' > 0$. Holder on $B(0, R_0 + t)$ gives

$$I'' \geq c(1+t)^{-3(p-1)} I^p. \tag{12.16}$$

负能量不需要初速度向外

聚焦幂 $f(z) = -|z|^{p-1}z$ 满足 $zf = (p+1)F$。取 $h = 0, g = A\psi$，A 足够大就使 $E = A^2\|D\psi\|_2^2/2 - A^{p+1}\|\psi\|_{p+1}^{p+1}/(p+1) < 0$。证明不依赖 $I'(0) > 0$。

Let v be the free wave with the same data. Positivity of the three-dimensional retarded kernel gives $u \geq v$, despite possible sign changes of v. Huygens' principle puts v in the shell $A_t = \{\max(t - R_0, 0) \leq |x| \leq t + R_0\}$, whose volume is at most $C(1 + t)^2$. Also $\int v = \int g + t \int h$. Consequently

$$c(1 + t) \leq \int_{A_t} u \leq \Big(\int |u|^p \Big)^{1/p} |A_t|^{1-1/p}, \qquad I'' \geq c(1 + t)^{2-p}.$$

Since $p < 3$, two integrations and the positive initial values yield $I(t) \geq c(1 + t)^{4-p}$ (after adjusting the constant for small t). This stronger lower bound comes from shell volume, not ball volume.

Choose $0 < \varepsilon < p-1$ so that $\mu = (p-1)^2 + \varepsilon(4-p) < 2$. Combining the lower bound with (12.16) gives $I'' \geq cI^{1+\varepsilon}(1 + t)^{-\mu}$. Multiply by $I' > 0$ and integrate. The product rule

$$(I^{2+\varepsilon}(1 + t)^{-\mu})' = (2 + \varepsilon)I^{1+\varepsilon}I'(1 + t)^{-\mu} - \mu I^{2+\varepsilon}(1 + t)^{-\mu-1}$$

shows $(I')^2 \geq cI^{2+\varepsilon}(1+t)^{-\mu} - C$. The positive term tends to infinity: its power lower bound has exponent $2(4 - p) - (p - 1)^2 > 0$ on the stated range. Thus for large t,

$$I' \geq cI^{1+\varepsilon/2}(1 + t)^{-\mu/2}, \qquad (I^{-\varepsilon/2})' \leq -c(1 + t)^{-\mu/2}.$$

Because $\mu/2 < 1$, the time integral on the right diverges. It makes the positive left quantity negative at a finite time, a contradiction. Scaling the initial data down changes constants and the possible lifetime, but does not remove this contradiction. □

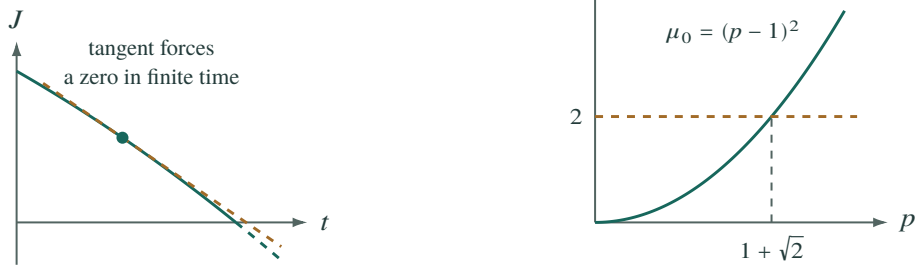


Diagram T12.4. Two different obstructions to a global solution On the left, a positive concave quantity with negative derivative lies below its tangent and must reach zero. The displayed curve $J = 1 - t - t^2/4$ is only an exact concavity model, not a PDE solution. On the right, $(p - 1)^2 < 2$ leaves room for the positive ε in the shell-based proof; the endpoint is excluded. Neither diagram asserts a quantitative lifespan formula for the PDE.

两个临界指数不可混用

这里 $1 + \sqrt{2}$ 来自 $(p - 1)^2 < 2$ 和球壳的 t^2 体积增长；三维退聚焦的能量临界 $p = 5$ 来自缩放。本节既没有证明 Strauss 临界端点，也没有讨论阈值以上的小数据全局理论。

12.6 Problems

These are abbreviated descriptions of all 18 source problems, followed by reconstructed solutions. Regularity, decay, and parameter qualifications are included where the source wording suppresses them.

1. Energy for a gradient-dependent principal part

For $u_{tt} - \operatorname{div} DL(Du) = 0$, use

$$E_L(t) = \int \{u_t^2/2 + L(Du) - L(0)\} dx.$$

The subtraction makes the integral finite for compactly supported u. Its derivative is $\int u_t u_{tt} + DL(Du) \cdot Du_t = \int u_t \{u_{tt} - \operatorname{div} DL(Du)\} = 0$. This needs no convexity of L for the identity; positivity or hyperbolicity would require further assumptions.

2. Klein–Gordon energy and equipartition

For $\Box u + m^2 u = 0$, take $m \geq 0$ and $E = \frac{1}{2} \int (u_t^2 + |Du|^2 + m^2 u^2)$. Multiplication by u_t proves conservation. For finite-energy data let $\omega(\xi) = \sqrt{|\xi|^2 + m^2}$ and use a unitary Fourier transform:

$$\hat{u} = \hat{g} \cos(\omega t) + \hat{h} \sin(\omega t)/\omega.$$

The difference between potential and kinetic terms is

$$\begin{aligned} D(t) &= \int (\omega^2 |\hat{u}|^2 - |\hat{u}_t|^2) d\xi \\ &= \int (\omega^2 |\hat{g}|^2 - |\hat{h}|^2) \cos(2\omega t) + 2\omega \operatorname{Re}(\hat{g}\overline{\hat{h}}) \sin(2\omega t)\, d\xi. \end{aligned}$$

Both amplitudes are L^1 by Cauchy–Schwarz and finite energy. In polar coordinates change the radial variable from r to $\omega = \sqrt{r^2 + m^2}$. The resulting amplitudes are L^1 on (m, ∞); the possible critical point $r = 0$ has no atom. Riemann–Lebesgue gives $D(t) \to 0$. Since potential plus kinetic is $2E$, each tends to E, including the claimed $\int (|Du|^2 + m^2 u^2) \to E$. A finite-energy approximation removes any auxiliary smoothness used for the Fourier calculation.

3. Descent and the one-dimensional massive kernel

Put $\overline{u}(x, y, t) = u(x, t) \cos(my)$. Then $(\partial_t^2 - \Delta_x - \partial_y^2)\overline{u} = \cos(my)(\Box u + m^2 u) = 0$. For $n = 1$, apply the two-dimensional Poisson formula and integrate out the additional variable. With $J_0(z) = \pi^{-1} \int_0^\pi \cos(z \sin\theta) d\theta$, the result is

$$\begin{aligned} u(x,t) = \frac{1}{2} \partial_t \int_{x-t}^{x+t} g(y) J_0\Big(m\sqrt{t^2 - (x-y)^2}\Big)\, dy \\ + \frac{1}{2} \int_{x-t}^{x+t} h(y) J_0\Big(m\sqrt{t^2 - (x-y)^2}\Big)\, dy. \end{aligned}$$

The integrated kernel identity is $\int_{-a}^{a} \cos(my)(a^2-y^2)^{-1/2}dy = \pi J_0(ma)$, obtained by $y = a\sin\theta$. If desired, differentiating the first integral gives endpoint term $\{g(x+t)+g(x-t)\}/2$ and interior kernel $-mtJ_1(m\sqrt{t^2-(x-y)^2})/(2\sqrt{t^2-(x-y)^2})$. Its endpoint limit is finite because $J_1(z) \sim z/2$. At $m = 0$ this reduces exactly to d'Alembert's formula.

4. Removing constant damping

For $\Box u + du_t = 0$, put $v = e^{dt/2}u$. Direct substitution gives $\Box v - (d^2/4)v = 0$. Thus the constant zero-order coefficient is negative when $d > 0$; the damping has not turned into a positive Klein–Gordon mass.

5. Phase speed and the domain of dependence

The plane wave $e^{i(x\cdot\xi-\omega t)}$ has dispersion $\omega^2 = |\xi|^2 + m^2$. For $m > 0$ its phase speed is $\omega/|\xi| > 1$, but its group velocity is $D_\xi\omega = \xi/\omega$, of magnitude less than one. A plane wave has no compact support and carries no newly created front. Causal propagation of a localized disturbance is the cone statement of §12.1, which remains true for the massive lower-order term. Group velocity describes narrow-band packets; the finite propagation proof is what controls an actual front. For $m = 0$ both speeds have magnitude one.

6. Uniform three-dimensional decay

Kirchhoff gives

$$u(x,t) = \frac{1}{4\pi}\int_{S^2}[g(x+t\omega) + tDg(x+t\omega)\cdot\omega + th(x+t\omega)]d\omega.$$

If the data are supported in $B(0,R)$ and $t \geq 2R$, the part of S^2 where $x+t\omega \in B(0,R)$ has area at most CR^2/t^2. This follows by intersecting the sphere of radius t centered at x with that fixed ball and using the spherical-cap area formula; if there is no intersection the integral is zero. The displayed formula is therefore bounded by C/t uniformly in x. For $t \leq 2R$, the same formula with the full sphere bounds u by a fixed constant, which is also at most a larger C/t.

7. Uniform two-dimensional decay

Differentiate the rescaled Poisson formula to write

$$u(x,t) = \frac{1}{2\pi}\int_{|x-y|<t} \frac{h(y) + \{g(y) + (y-x)\cdot Dg(y)\}/t}{\sqrt{t^2-|x-y|^2}}dy.$$

For t larger than a fixed multiple of the support radius, its absolute value is bounded by a data constant times $J = \int_{B_R \cap B(x,t)} (t^2 - |x-y|^2)^{-1/2} dy$. On $|x-y| \le t/2$ the contribution is C_R/t. For the remaining radii use coarea about x. The length of a circle of radius $r > t/2$ intersected with B_R is at most CR; the possible radii form an interval of length at most $2R$. Since the kernel increases toward $r = t$, this contribution is at most

$$CR \int_{\max(t-2R,\,t/2)}^{t} \frac{dr}{\sqrt{t^2 - r^2}} \le C_R t^{-1/2}.$$

The bounded small-time strip follows directly from the rescaled Poisson formula and the smooth data. Hence $|u(x,t)| \le Ct^{-1/2}$ for all $x, t > 0$. Unlike the three-dimensional case, the interior of the light cone generally carries a tail.

8. Bounded displacement data can amplify

For the given radial g^ε, $0 \le g^\varepsilon \le 1$ and $g^\varepsilon(r) \to \mathbf{1}_{(1,3)}(r)$ a.e. as $\varepsilon \downarrow 0$. At $x = 0$ and fixed $t > 3$, Poisson gives

$$u^\varepsilon(0,t) = -t \int_1^3 \frac{r g^\varepsilon(r)}{(t^2 - r^2)^{3/2}} dr \longrightarrow t \left\{ \frac{1}{\sqrt{t^2 - 1}} - \frac{1}{\sqrt{t^2 - 9}} \right\}.$$

Dominated convergence is valid for each such fixed t. The limit tends to $-\infty$ as $t \downarrow 3$. Given any M, first choose $3 < t < 4$ making its magnitude exceed $2M$, then take ε small enough. This proves that the supremum in the problem tends to infinity. No maximum principle for wave displacement follows from a bound on g alone.

9. Hyperbolic Kelvin transform

Let $d = n + 1$, $X = (x,t)$, $q = |x|^2 - t^2 \neq 0$, $Y = X/q$, $a = (d-2)/2$ and $w(X) = |q|^{-a}$. For the Lorentz matrix $G = \operatorname{diag}(-1, \dots, -1, 1)$, direct differentiation gives

$$DY\, G\, (DY)^T = q^{-2} G, \qquad \Box w = 0, \qquad w \Box Y^j + 2\langle Dw, DY^j \rangle_G = 0.$$

For example $\Box q = -2d$, $\langle Dq, Dq \rangle_G = -4q$, $\Box Y^j = 2(d-2)X^j/q^2$, and $\langle Dw, DY^j \rangle_G = -2awX^j/q^2$. The product and chain rules therefore cancel every first-derivative term and leave

$$\Box\{w\, u(Y)\} = |q|^{-(d+2)/2} (\Box u)(Y).$$

It follows that the transform preserves free waves on each region where $q \neq 0$. Absolute values allow both signs of q; no extension through the characteristic cone is asserted.

10. Compatibility of the sine–Gordon system

Set $A = (u+v)/2$, $B = (u-v)/2$. The given equations are $B_t = a\sin A$, $A_x = a^{-1}\sin B$. Differentiating gives $B_{xt} = \cos A \sin B$ and $A_{xt} = \cos B \sin A$. Adding and subtracting yield $u_{xt} = \sin(A+B) = \sin u$ and $v_{xt} = \sin(A-B) = \sin v$. These x, t are characteristic coordinates. With new variables $T = x + t$, $X = x - t$, the mixed derivative becomes $\partial_x \partial_t = \partial_T^2 - \partial_X^2$. Thus the equation has the usual wave-operator form in (X, T), with no missing factor when these summed coordinates are used.

11. A Bäcklund kink from the zero solution

With $v = 0$, the equations become $u_t = 2a\sin(u/2)$, $u_x = 2a^{-1}\sin(u/2)$. They imply $u = U(s)$, $s = at + x/a$, and $U' = 2\sin(U/2)$. Since $(\tan(U/4))' = \tan(U/4)$, nonconstant solutions are

$$u(x,t) = 4\arctan(Ce^{at+x/a}), \qquad C \neq 0,$$

with compatible 4π shifts; the constant solutions $u = 2\pi k$ are included separately. Differentiation gives $U'' = \sin U$, so $u_{xt} = U'' = \sin u$. For $a > 0, C > 0$ the profile connects 0 to 2π as s increases. The parameter C translates the kink.

12. The Sobolev product estimate

For $b_j = |\beta_j|$, the proof of (12.5) gives $D^{\beta_j} u_j \in L^{p_j}$ for every inverse exponent between $b_j/(2k)$ and $1/2$, when $k > n/2$. Choose these inverse exponents with sum exactly $1/2$; the endpoint sums bracket $1/2$ because $\sum b_j \le k$. Holder then gives $\| \prod D^{\beta_j} u_j \|_2 \le C \prod \|u_j\|_{H^k}$. This handles also zero multiindices and the case of a derivative of order exactly k, without using the false endpoint embedding $H^{n/2} \hookrightarrow L^\infty$.

13. Sphere-valued wave maps

Since $|u|^2 = 1$, differentiation gives $u \cdot u_{tt} = -|u_t|^2$ and $u \cdot \Delta u = -|Du|^2$. The normal space of the sphere is spanned by u itself. Thus $\Box u = \gamma u$ with $\gamma = u \cdot \Box u = |Du|^2 - |u_t|^2$, as asserted. This is a genuine derivative nonlinearity, so the scalar potential energy theory is not directly applicable.

14. Wave-map energy and a support correction

A unit-length map cannot itself vanish outside a compact set. Interpret the source's support wording as u equal to a fixed unit vector outside a compact set, or impose decay on Du, u_t sufficient for integration by parts.

Then

$$\frac{d}{dt}\frac{1}{2}\int(|u_t|^2+|Du|^2)=\int u_t\cdot\Box u=\int(|Du|^2-|u_t|^2)u_t\cdot u=0,$$

because $u_t\cdot u=0$. This states the nonvacuous finite-energy version of the exercise.

15. An elementary small-energy quintic proof

Modify the average estimate in (12.7) using $\int_{B_t}|g|\le Ct^{5/2}\|g\|_6$. Then

$$|\overline{w}|\le Ct^{-1/2}(E^{1/2}+\|g\|_6),\qquad \|w/r\|_2\le C(E^{1/2}+E^{1/6}),$$

since $\|g\|_6^6\le 6E$. Cone flux also gives $\int w^6\le 6E$. For $M(\tau)=\sup_{t\le\tau,x}|u(x,t)|$,

$$\int\frac{|w|^5}{r}\le\|w/r\|_2\left(\int|w|^8\right)^{1/2}\le C(E+E^{2/3})M(\tau).$$

The retarded formula gives $M(\tau)\le\|v\|_\infty+C(E+E^{2/3})M(\tau)$. Choose a universal small initial-energy threshold making the last coefficient at most $1/2$. On every finite strip the free wave is bounded, so the amplitude is bounded and the smooth solution continues globally. This argument needs neither the Morawetz identity nor Strichartz, and makes no smallness assumption on a high Sobolev norm of the data.

16. Galilean invariance for NLS

For $iu_t+\Delta u=f(|u|^2)u$, set $\theta=\xi\cdot x/2-|\xi|^2t/4$, $y=x-\xi t$, $w=e^{i\theta}u(y,t)$. Then

$$iw_t=e^{i\theta}(|\xi|^2u/4+iu_t-i\xi\cdot Du),$$
$$\Delta w=e^{i\theta}(\Delta u+i\xi\cdot Du-|\xi|^2u/4).$$

The extra terms cancel, and $|w|=|u(y,t)|$ makes the nonlinear term transform with the same phase. This proves the stated invariance.

17. NLS mass, energy, and momentum

Assume the solution and the needed derivatives decay rapidly. Let $\rho=|u|^2$, $F(0)=0$, $F'=f$, with f real. The mass identity follows from $\rho_t=-2\operatorname{div}\operatorname{Im}(\overline{u}Du)$. For the energy $\mathscr{H}=\int(|Du|^2+F(\rho))$,

$$\mathscr{H}'=2\operatorname{Re}\int(-\Delta\overline{u}+f(\rho)\overline{u})u_t=2\operatorname{Re}\int(-i\overline{u}_t)u_t=0.$$

For $P_j=\operatorname{Im}\int\overline{u}\,u_{x_j}$, integration by parts in one of the two time-derivative terms gives

$$P_j'=2\operatorname{Im}\int\overline{u}_tu_{x_j}=-2\operatorname{Re}\int\Delta\overline{u}\,u_{x_j}+2\operatorname{Re}\int f(\rho)\overline{u}\,u_{x_j}=0.$$

球面值映射不能紧支为零

若 $|u|=1$，则 $u\not\equiv 0$ 于任何外部区域。题 14 的紧支应理解成 $u-u_\infty$ 紧支或导数充分衰减。能量只积分 Du,u_t，不积分处处为 1 的 $|u|^2$。

The last two integrals are respectively integrals of $\partial_j |Du|^2$ and $\partial_j F(\rho)$, hence vanish. This agrees with the source's momentum integrand $(\overline{u}Du - uD\overline{u})/(2i)$.

18. NLS virial identity and focusing cubic blow-up

In addition assume finite variance $V(t) = \int |x|^2 |u|^2$ and sufficient decay for the weighted integrations. The mass continuity identity gives $V' = 4\,\mathrm{Im} \int x \cdot \overline{u} Du$. Differentiate this expression, substitute $u_t = i\Delta u - if(\rho)u$, and integrate twice in the Laplacian terms. They give $2\int |Du|^2$ in the derivative of $V'/4$. The nonlinear terms give

$$-\int x \cdot \rho Df(\rho) = -\int x \cdot D\{\rho f(\rho) - F(\rho)\} = n \int \{\rho f(\rho) - F(\rho)\}.$$

Thus

$$V'' = 8\int |Du|^2 + 4n \int \{\rho f(\rho) - F(\rho)\}. \tag{12.17}$$

For focusing cubic NLS, $f(\rho) = -\rho$, $F = -\rho^2/2$ and $\mathcal{H} = K - P/2$, with $K = \int |Du|^2$, $P = \int |u|^4$. Hence $V'' = 8K - 2nP = 8\mathcal{H} + (4 - 2n)P \le 8\mathcal{H}(0) < 0$ when $n \ge 2$ and the initial energy is negative. Integrating gives $0 \le V(t) \le V(0) + V'(0)t + 4\mathcal{H}(0)t^2$, which fails at a finite time. This proves nonexistence of a global solution in the stated regularity and finite-variance class. The variance hypothesis is essential to this particular proof; negative energy alone is not a license to differentiate an infinite weighted integral.

12.7 References

Source attributions

Evans acknowledges D. Tataru's comments. His source for §12.2 and parts of §12.3 is Shatah–Struwe [S-S]. He attributes the sign-condition weak-existence argument to I. Segal, Bull. Soc. Math. France 91 (1963), 129–135, and the three-dimensional/subcritical arguments to K. Jörgens, Math. Z. 77 (1961), 295–308. The critical flux and global-existence theorems are attributed to M. Grillakis, Ann. of Math. 132 (1990), 485–509, with Smith–Sogge simplifications, and reference to Sogge [So] and Shatah–Struwe. Strauss [St1] and John [J3] are cited for blow-up. Problem 8 is credited to Protter–Weinberger [P-W]; small-energy quintic existence to J. Rauch; J. Holmer is thanked for help with Problems 16–18. The cubic NLS blow-up exercise cites R. Glassey, J. Math. Physics 18 (1977), 1794–1797. These are source attributions, not a claim to have separately consulted every listed publication.

Proof boundaries and editorial changes

The chapter uses the earlier linear wave, Sobolev, Hardy, Rellich, Gronwall, Banach fixed-point, Fourier, and measure-theoretic tools. The product exponent allocation and completeness of the strong ball in the weak metric are supplied. Sign-condition weak existence uses space-time uniform integrability; its energy inequality is distinguished from smooth conservation and supplies only finite-strip L^2 bounds. For noninteger powers, global bounded finite-energy mild solutions are asserted instead of unqualified smoothness. Sphere-valued maps are interpreted as constant outside a compact set, not zero there.

The homogeneous $(4, 12)$ three-dimensional Strichartz estimate is an explicit external theorem. Its dispersive proof is not supplied; the energy and Duhamel parts, localization by cone-supported forcing, Morawetz identity, singular-top limit, nonconcentration, cone enlargement, finite cover, and continuation are expanded. No solution at the candidate singular time is assumed in advance. The critical proof does not claim a general supercritical theorem or scattering. The focusing small-data proof covers $1 < p < 1 + \sqrt{2}$, not the critical endpoint or the range above it. The NLS virial argument assumes finite variance and enough decay. All 18 solutions, Chinese notes, and five diagrams are editorial reconstructions. Source comparison, symbolic checks, compilation, and visual review are separate from independent mathematical review, which has not been performed.